

Python Programming

1: Design a Python application that creates two separate threads named Even and Odd.

- The **Even** thread should display the **first 10 even numbers**.
- The **Odd** thread should display the **first 10 odd numbers**.
- Both threads should execute independently using the **threading** module.
- Ensure proper thread creation and execution.

2: Design a Python application that creates two threads named EvenFactor and OddFactor.

- Both threads should accept **one integer number as a parameter**.
- The **EvenFactor** thread should:
 - Identify all even factors of the given number.
 - Calculate and display the **sum of even factors**.
- The **OddFactor** thread should:
 - Identify all odd factors of the given number.
 - Calculate and display the **sum of odd factors**.
- After both threads complete execution, the **main thread should display the message: "Exit from main"**

3: Design a Python application that creates two threads named EvenList and OddList.

- Both threads should accept a **list of integers** as input.
- The **EvenList** thread should:
 - Extract all even elements from the list.
 - Calculate and display their **sum**.
- The **OddList** thread should:
 - Extract all odd elements from the list.
 - Calculate and display their **sum**.
- Threads should run concurrently.

4: Design a Python application that creates three threads named Small, Capital, and Digits.

- All threads should accept a **string as input**.
- The **Small** thread should count and display the **number of lowercase characters**.
- The **Capital** thread should count and display the **number of uppercase characters**.
- The **Digits** thread should count and display the **number of numeric digits**.
- Each thread must also display:
 - **Thread ID**
 - **Thread Name**

5: Design a Python application that creates two threads named Thread1 and Thread2.

- **Thread1** should display numbers from **1 to 50**.
- **Thread2** should display numbers from **50 to 1** in reverse order.
- Ensure that:
 - **Thread2 starts execution only after Thread1 has completed.**
- Use appropriate thread synchronizatio